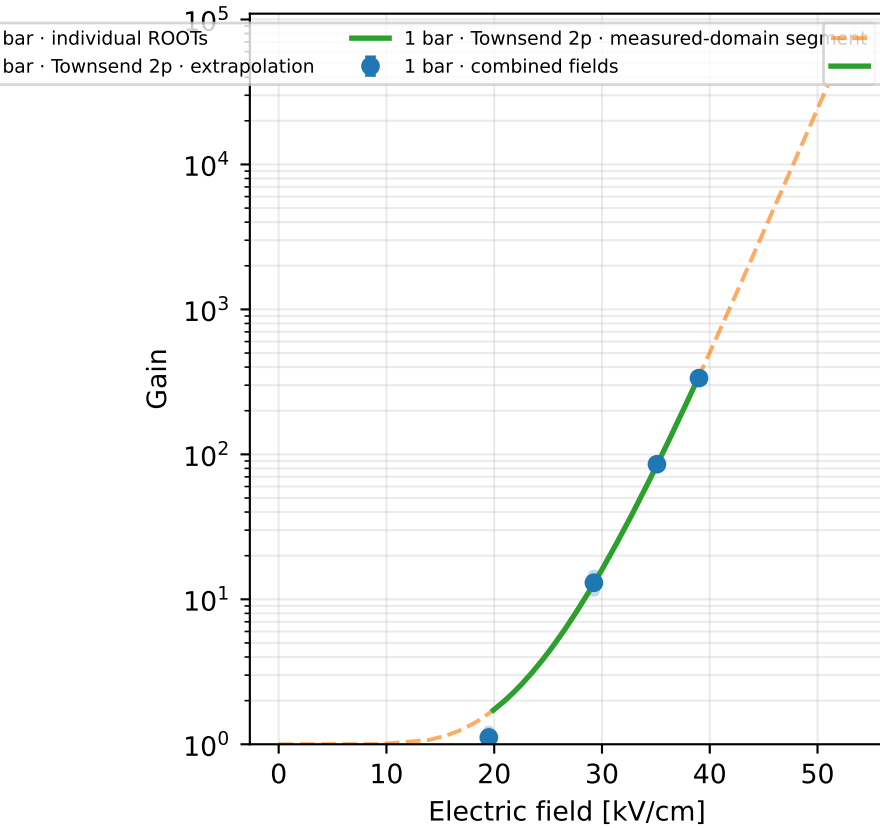
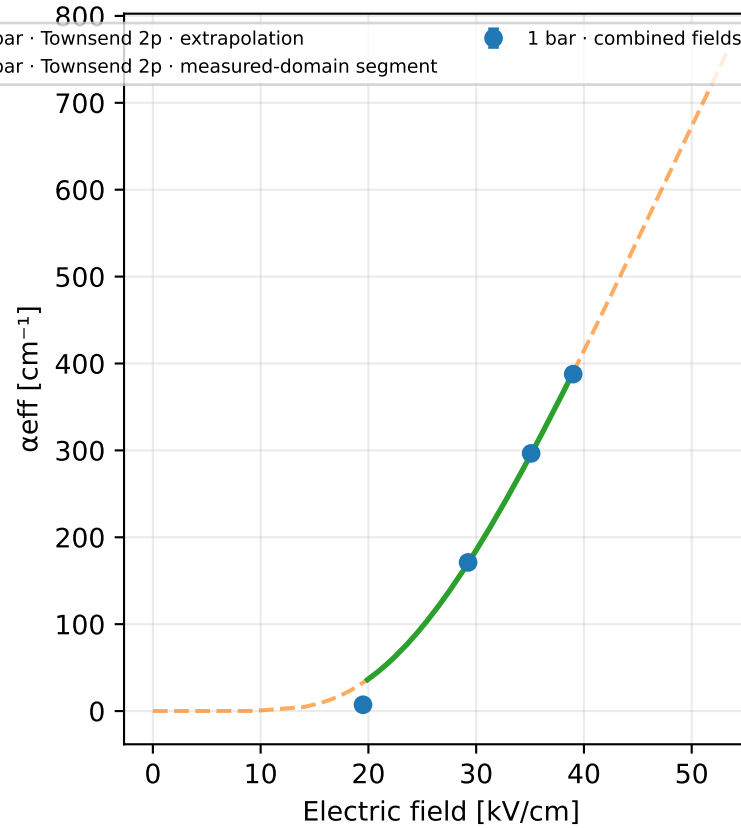


ArCF4Iso · Ar 78 % / CF₄ 20 % / Iso 2 % · gap 0.15 mm

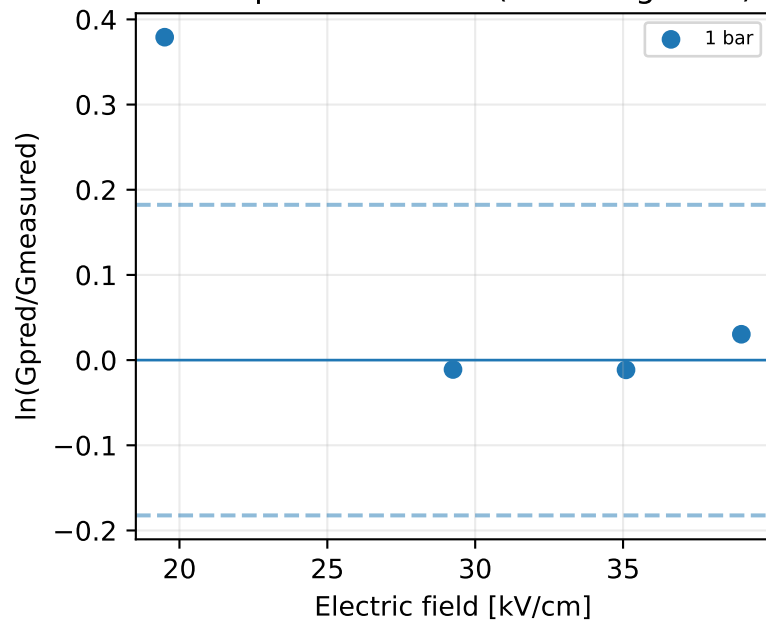
Gain vs electric field



Effective Townsend coefficient vs electric field



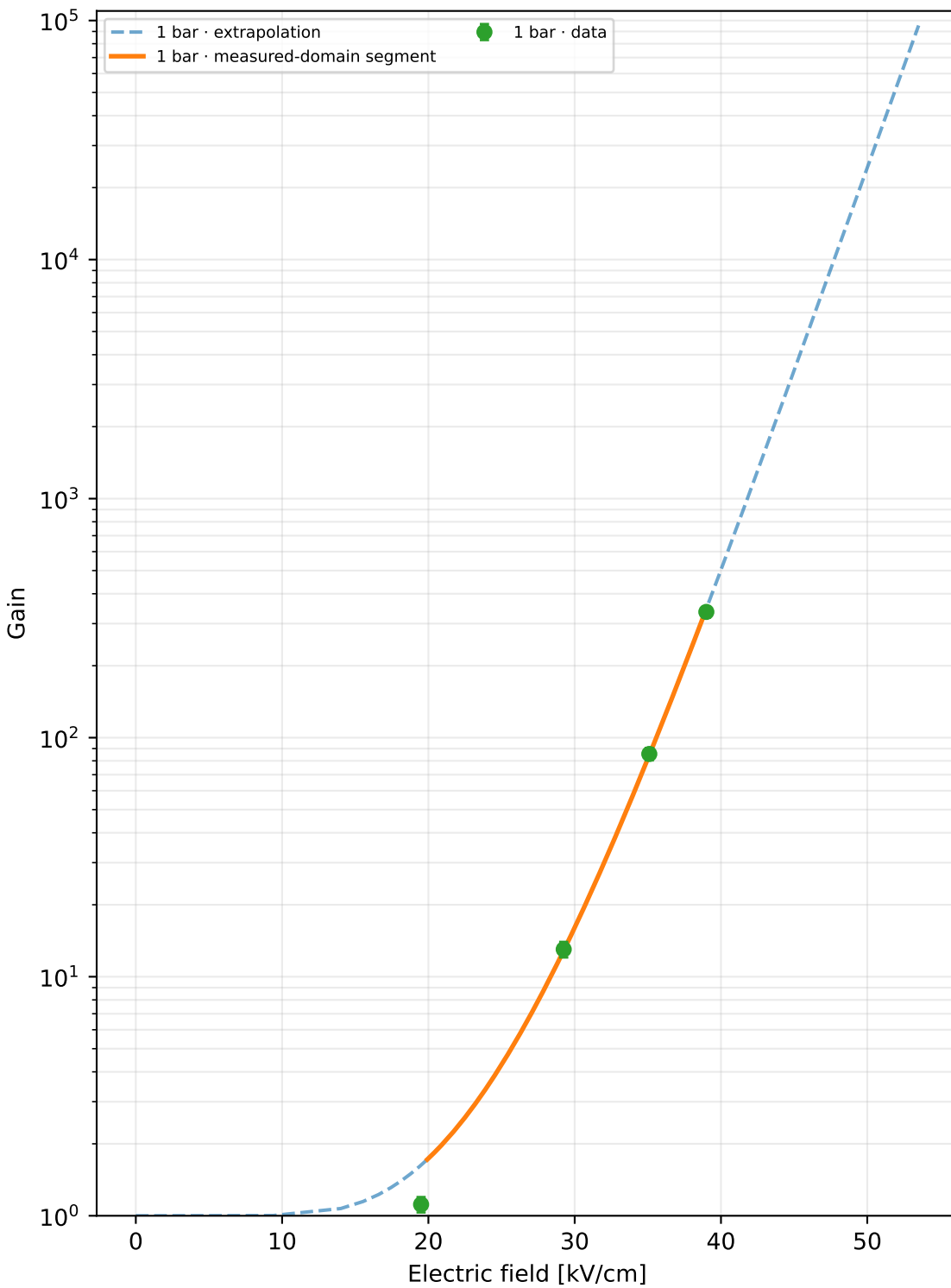
Gain-space residuals ($\pm \ln 1.2$ guides)



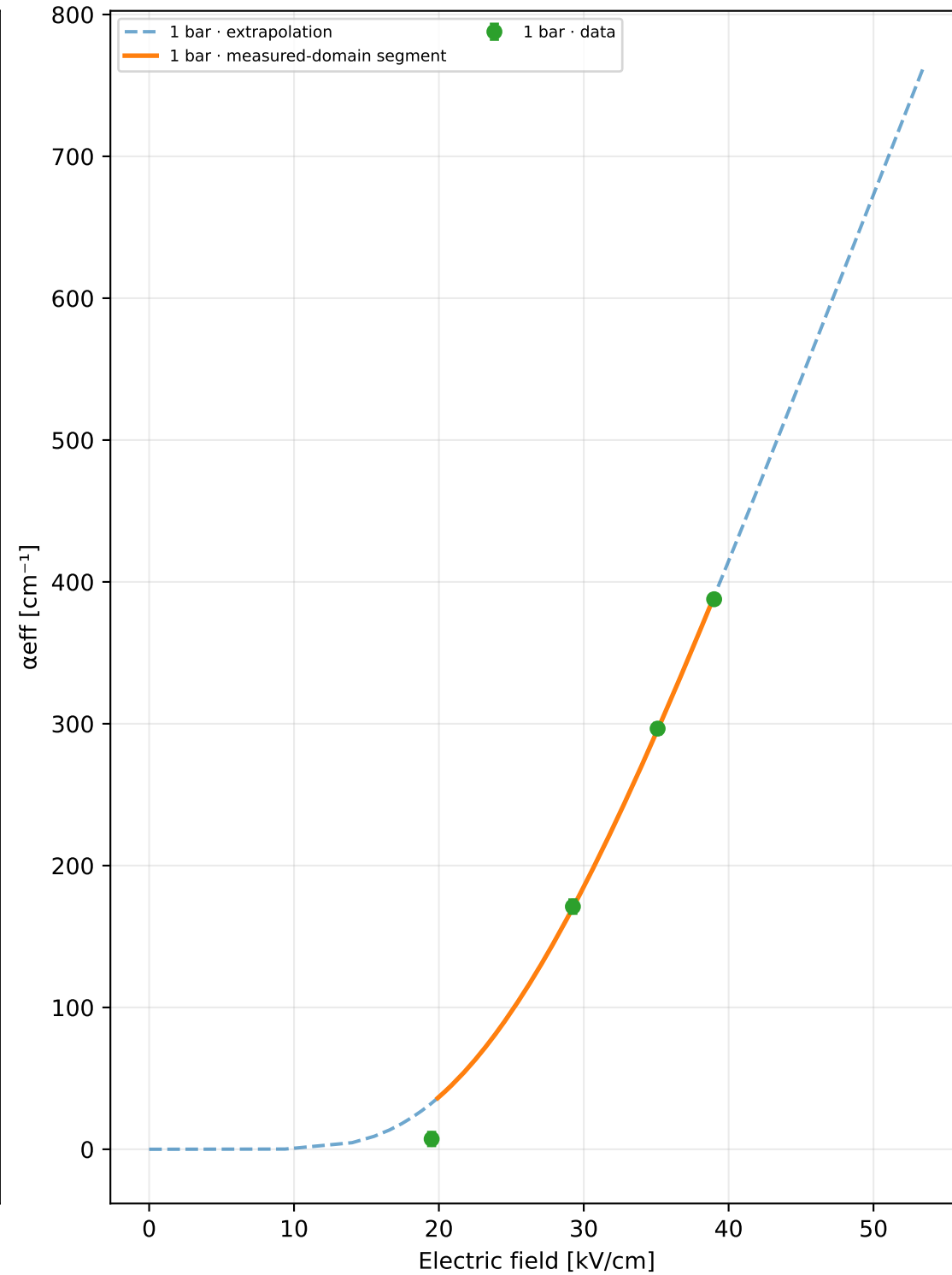
ROOT runs: 8
 Unique physical points: 4
 Pressures: 1 bar
 Selected model: Townsend 2p
 Status: rejected_poor_cross_validation
 Training RMSE: $\times 2.117$
 Cross-validation median: unavailable
 Cross-validation worst: unavailable
 Reduced-field range: 19.5–39 kV cm⁻¹ bar⁻¹
 A=4670.6, B=96.855, m=0, n=1
 Covariance condition: 725
 Warnings: poor_cross_validation, large_cross_validation_outlier, large_training_resi
 1 bar display limit: G=9.54e+04 (local inversion limit)

Extrapolation capped at $G=1e+07$ or at the local model limit

Gain vs electric field



α_{eff} vs electric field



p [bar]	E [kV/cm]	E/p	Gain	σ Gain	α_{eff} [cm ⁻¹]	npe	runs
1	19.5	19.5	1.115	0.083	7.25696	200	2
1	29.25	29.25	13.015	0.98	171.074	200	2
1	35.1	35.1	85.485	5.14	296.556	200	2
1	39	39	335.78	18.2	387.764	200	2

Progressive model selection

The higher-order model is selected only when new points improve out-of-sample prediction.

Candidate	k	AICc	CV median	CV worst	usable	warnings
Townsend 2p	2	13.7	—	—	no	poor cross validation, large cross validation outlier, large training residuals